

Dumpsite area, perimeter and shape from traced outlines

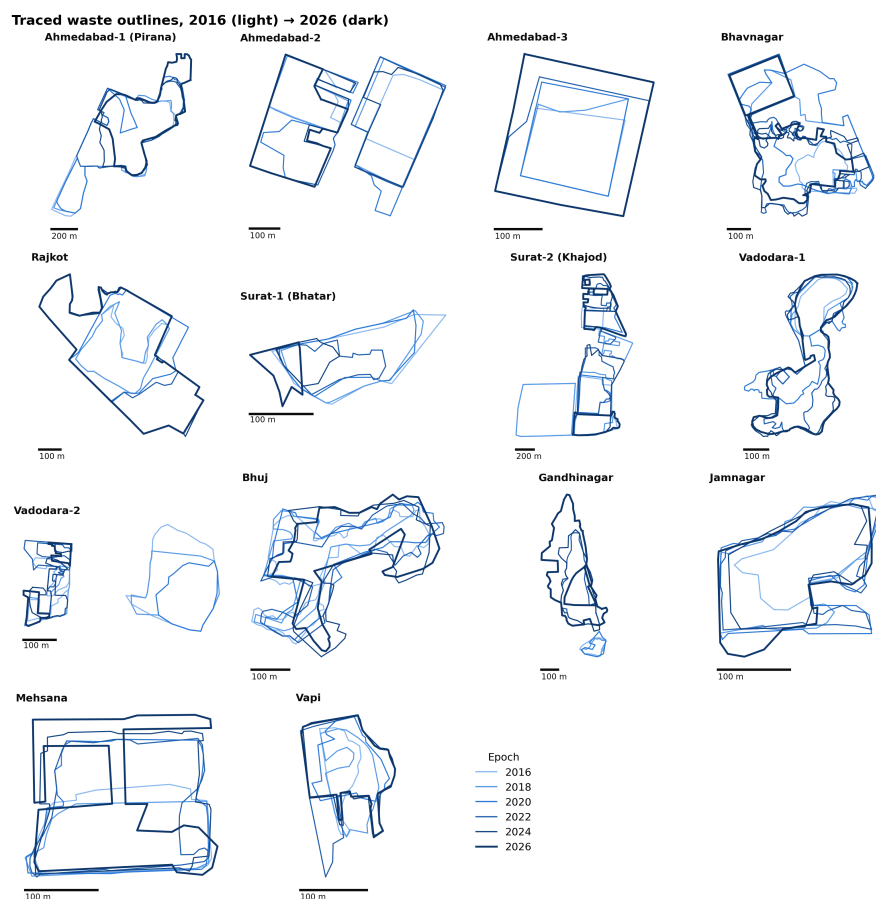
Method, data sources and a step-by-step guide to reproduce every figure in the repository

Gujarat dumpsite RS-GIS study (14 sites, 10 cities, six epochs 2016-2026) · Integrated Waste Management and Circular Economy, mid-semester project

Repository: [wre9-tesh/waste-managment-project-1](#) · Scripts: `scripts/final_analysis.py` (Step 1) and `scripts/final_figures.py` (Step 2) · Re-run and checked: 26 September 2026

Master outlines: Locations of landfills in gujurat completer.kml (repository root; 122 placemarks: 103 traced outlines + 19 pins). These are the same polygons the volume guide (volume_final_26Sep) uses.

Use of this document. This is a working and reproducibility record prepared with Claude for your own use: methods, commands, checks and numbers. Under the course rule on AI-generated content, do not copy its sentences into the slides or report; write those yourself. Figures and numbers are analysis outputs you can use.



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Reading order to reproduce the figures: Sections 4 → 5 → 6 → 7, then the manual check in Section 9. Section 8 explains each figure; Appendix A has the 14 per-site outline maps.

1. What was done and what came out

Waste outlines were traced by hand in Google Earth Pro historical imagery for 14 dumpsites in 10 Gujarat cities at six epochs (2016, 2018, 2020, 2022, 2024, 2026) and saved in one KML file. Two Python scripts turn that KML into every table and figure in the repository:

Step	Script	Reads	Writes
1	scripts/final_analysis.py	Master KML	tables/final_results.xlsx and 6 CSVs; data/outlines_by_site_epoch.geojson, data/outlines_by_part.geojson, data/results.json
2	scripts/final_figures.py	data/results.json, data/gujarat_boundary.geojson	figures/fig01-fig06, fig08-fig13 (12 PNGs) and figures/site_outlines/ (14 PNGs), 200 dpi

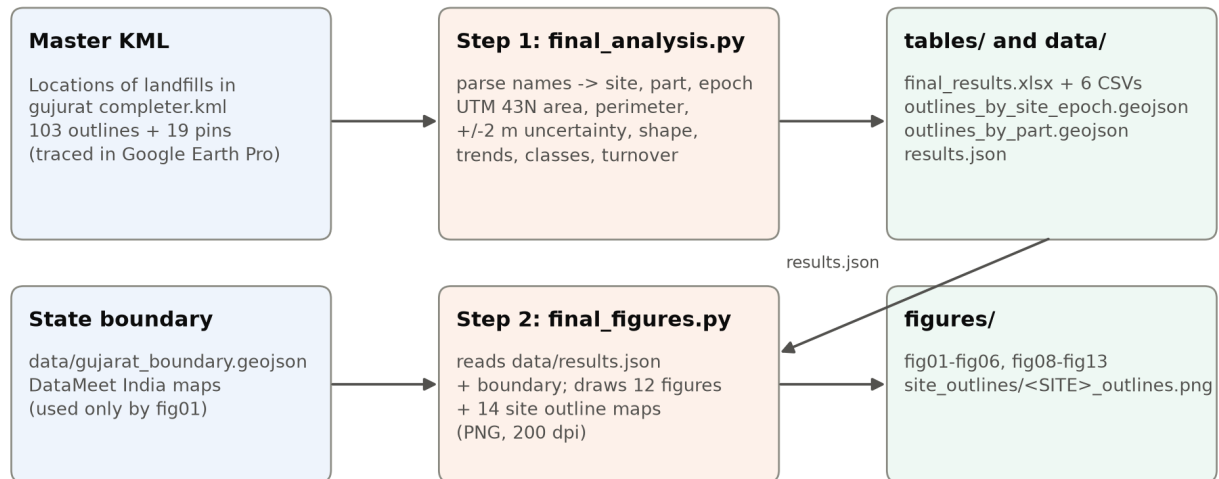


Figure 1.1. Workflow. Inputs on the left, the two scripts in the middle, outputs on the right. Step 2 never reads the KML; it only reads results.json (and the state boundary for the map), so the figures always match the tables.

1.1 Headline numbers

ID	Site	2016 (ha)	2026 (ha)	Net (ha)	Net (%)	Largest (year)	Class
AHM-01	Ahmedabad LF-1 (Pirana)	38.5	29.9	-8.6	-22%	38.7 (2020)	Contracting
AHM-02	Ahmedabad LF-2	8.5	8.2	-0.3	-3%	18.0 (2022)	Rise and fall
AHM-03	Ahmedabad LF-3	3.1	8.5	+5.4	+175%	8.5 (2024)	Expanding
BHV-01	Bhavnagar LF-1	3.7	14.8	+11.1	+299%	19.7 (2020)	Rise and fall
RAJ-01	Rajkot LF-1	7.3	28.1	+20.8	+285%	28.3 (2024)	Expanding
SUR-01	Surat LF-1 (Bhatar)	2.0	0.4	-1.6	-79%	2.0 (2016)	Contracting
SUR-02	Surat LF-2 (Khajod)	68.7	31.2	-37.5	-55%	68.7 (2016)	Cleared then regrowing
VAD-01	Vadodara LF-1	3.0	12.9	+9.9	+326%	12.9 (2026)	Expanding
VAD-02	Vadodara LF-2	7.0	0.7	-6.2	-89%	7.0 (2016)	Contracting
BHJ-01	Bhuj LF-1	5.0	6.7	+1.7	+34%	7.7 (2024)	Expanding
GNR-01	Gandhinagar LF-1	1.1	12.6	+11.5	+1033%	12.6 (2026)	Expanding
JAM-01	Jamnagar LF-1	1.6	2.8	+1.2	+80%	2.9 (2020)	Expanding
MEH-01	Mehsana LF-1	2.6	2.8	+0.2	+8%	4.3 (2024)	Rise and fall
VAP-01	Vapi LF-1	0.5	1.4	+0.9	+167%	1.7 (2022)	Expanding
	Five most populous cities (9 sites)	141.7	134.8	-6.9	-5%	167.9 (2024)	
	Other cities (5 sites)	10.7	26.2	+15.5	+144%	26.2 (2026)	
	All 14 sites	152.5	161.0	+8.6	+6%	192.1 (2024)	

Class rules are in Section 6.5. Blue/orange colouring in the figures = five most populous cities (Ahmedabad, Surat, Vadodara, Rajkot, Bhavnagar) / other cities.

1.2 Key analysis points (for your own wording)

- Total, all 14 sites: 152.5 ha (2016) → 142.7 (2020) → 192.1 (2024, largest) → 161.0 ha (2026) = +6% net.
- Five most populous cities (9 sites): 141.7 → 134.8 ha (−5%); without Khajod 73.0 → 103.6 ha (+42%). Other cities (5 sites): 10.7 → 26.2 ha (+144%).
- Rajkot +20.8, Gandhinagar +11.5, Bhavnagar +11.1 and Vadodara LF-1 +9.9 ha together are 85% of all net gain (62.7 ha summed over the 9 growing sites).
- Largest declines: Khajod −37.5 ha (−55%), Pirana −8.6 ha (−22%), Vadodara LF-2 −6.2 ha (−89%), Surat-Bhatar −1.6 ha (−79%).
- Khajod: 68.7 → 19.0 ha in 2020 (part (a), 36.8 ha, cleared) → 53.5 in 2024 → 31.2 in 2026 (class "Cleared then regrowing").
- Land turnover: 97.7 ha newly covered since 2016; 142.4 ha held waste at some point but is not under waste in 2026.
- Mann-Kendall $p < 0.05$ (exact, $n = 6$, descriptive only): Ahmedabad LF-3, Gandhinagar, Rajkot, Vadodara LF-1 increasing; Surat-Bhatar, Vadodara LF-2 decreasing.
- Compactness fell sharply at Gandhinagar (0.80 → 0.17), Mehsana (0.73 → 0.15), Vadodara LF-1 (0.70 → 0.31) and Bhavnagar (0.44 → 0.23).

2. Folder layout

Everything sits in the repository folder. The scripts find their inputs relative to their own location (BASE = the folder above scripts/), so the folder can be copied anywhere.

Path (inside the repository)	Contents
Locations of landfills in gujarat completer.kml	MASTER input: all traced outlines and location pins (Section 5)
scripts/final_analysis.py	Step 1: reads the KML, computes all metrics, writes tables/ and data/
scripts/final_figures.py	Step 2: reads data/results.json, draws all figures
scripts/requirements.txt	Python packages for both steps
scripts/clip_dem_tiles.py	Not used here (volume work: clips DEM tiles per site)
data/gujarat_boundary.geojson	Gujarat state outline (input, used only by fig01)
data/outlines_by_site_epoch.geojson	Output: one polygon per site and epoch, parts merged (84 features, WGS84)
data/outlines_by_part.geojson	Output: one polygon per site, epoch and part (103 features, WGS84)
data/results.json	Output: every number the figures use, plus outline shapes and marker positions
tables/final_results.xlsx	Output: sheets site_summary, site_by_epoch, parts_by_epoch, city_totals, group_totals, transitions, notes
tables/*.csv	Output: the same six tables as CSV (no notes sheet)
figures/fig*.png, figures/site_outlines/*.png	Output: 12 figures + 14 per-site outline maps (200 dpi PNG)
gujarat_dumpsite_footprints_final.html	Results web page. Made separately from results.json; not written by these scripts
README.md, context.md	Figure list and key numbers; project log
docs/	This document

3. Data used

3.1 Sources

Dataset	Used for	Dates	Access
Google Earth Pro historical imagery (Maxar, Airbus/CNES and others as credited in Google Earth)	Background for hand tracing of every outline	Imagery 2016-2026; one image per site per epoch (2025 image used for 2024 at 5 sites)	Free desktop application (Google Earth Pro, "Historical imagery" clock)
Master KML (your tracing)	All areas, perimeters, shapes, trends and outline maps	2016, 2018, 2020, 2022, 2024, 2026	Repository root
DataMeet Community Maps, India states	Gujarat outline in the study-area map (fig01) only	n/a	github.com/datameet/maps (open data); Gujarat feature saved as data/gujarat_boundary.geojson

No other data enter the figures. City grouping (five most populous cities vs other cities) is a fixed list in the script (Appendix C).

3.2 Tracing rules (recorded in context.md and tracing_protocol.md)

- Manual tracing in Google Earth Pro historical imagery (automatic Sentinel-2 outlines were tried and rejected: at 10 m, cleared or biomined bare ground looks like waste).
- Name of each outline: <City> LF-<n> - <year>; parts as (a), (b), (c); year = image year; a 2025 image is used as the 2024 epoch.
- A pin named with a year and "remediated" or "NONE" means no waste at that site part in that year (area 0).
- Leachate pools are excluded (water ponding on top of the mound stays in). At biomining sites, waste still present counts; trommels, sheds, separated output stockpiles and cleared ground do not.

3.3 Epochs and imagery

Epoch	Image year used	Notes
2016, 2018, 2020, 2022	Same year	All 14 sites
2024	2024, or 2025 where no usable 2024 image	2025 used at Ahmedabad LF-1, LF-2(a), LF-3, Bhuj and Vapi (the script logs each case)
2026	2026	All 14 sites

4. Software setup (once)

1. Install Python 3.10 or newer from python.org (tick *Add python.exe to PATH* during install).
2. Get the repository: download it from GitHub (Code ► Download ZIP) and unzip it, or `git clone https://github.com/wre9-tesh/waste-managment-project-1`.

3. Open Command Prompt and go to the scripts folder (adjust the path to where your copy is):
`cd /d "D:\Coursework\Waste and circular economy\Project 1\waste-managment-project-1\scripts"`

1. Install the packages (all have ready-made Windows wheels; no GDAL install needed):
`python -m pip install -r requirements.txt`

1. Check the install:

```
python -c "import numpy, pandas, shapely, pyproj, matplotlib, openpyxl; print('ok')"
```

Versions used for the check run in this document (Linux). Newer or older versions give the same tables; figure PNGs can differ by a few pixels in size because font rendering and tight cropping change slightly between matplotlib versions (Section 10).

Package	Version used in the check run	Used for
python	3.11.15	
shapely	2.1.2	Polygons, union, buffer, simplify, difference
pyproj	3.7.2	WGS84 → UTM 43N (EPSG:32643) and back
pandas	3.0.6	Tables, Excel/CSV output
openpyxl	3.1.5	Excel writer used by pandas
numpy	2.4.6	Figures
matplotlib	3.11.2	Figures (font DejaVu Sans ships with it)

Optional for the manual check (Section 9): Google Earth Pro and/or QGIS 3.x.

5. Step 0 - the master KML and how it is read

5.1 What is in the file

Locations of landfills in `gujurat completer.kml` must sit in the repository root with exactly this name (the script path is fixed). It holds 15 folders (one or more per city) with 122 placemarks:

103 polygons (traced outlines) and **19 point pins** (site locations, plus two "remediated" pins that record a zero).

Site	Name in KML (example)	Parts	Polygons
AHM-01	ahmedabad LF - 1(2016)	one	6
AHM-02	Ahmedabad LF - 2(a) (2016)	(a), (b)	11
AHM-03	Ahmedabad LF -3 (2025)	one	6
BHV-01	Bhavnagar LF -2026	one	6
RAJ-01	Rajkot LF - 2016	one	6
SUR-01	Surat LF - 1 (2016) (Bhatar Old Garbage Dumping Site)	one	6
SUR-02	Surat LF - 2(c) 2026	(a), (b), (c)	14
VAD-01	Vadodara LF - 1(2016)	one	6
VAD-02	Vadodara LF - 2(a + b) 2018	old dump, (a), (b), (a+b)	12
BHJ-01	Bhuj LF - 2016	one	6
GNR-01	Gandhinagar LF -2016	one	6
JAM-01	Jamnagar LF - 2016	one	6
MEH-01	Mehsana LF - 2016	one	6
VAP-01	Vapi-LF - 2025	one	6
	Total		103

5.2 How a placemark name is turned into site, part and epoch

Function `parse()` in `final_analysis.py`. The name is lower-cased and the typo "vadoara" is corrected; then:

Item	Rule	Example → result
City	First word of the name; must be one of the 10 study cities, else the placemark is ignored	"Vapi-LF - 2025" → vapi
Landfill number	Digit after "LF" (spaces and "-" allowed). No digit → LF-1	"Bhuj LF - 2016" → LF-1 → BHJ-01
Part	Letters in brackets right after the LF number: (a), (b), (c), (a + b) → a+b. None → "main"	"Surat LF - 2(c) 2026" → part c
Year	First 20xx in the name. 2025 → epoch 2024	"Ahmedabad LF -3 (2025)" → 2024
Zero flag	Name contains remed / remid / none	"Surat LF - 2(a) 2020 Remediated" → part a = 0 in 2020
Site ID	City code + LF number: AHM, SUR, VAD, RAJ, BHV, GNR, JAM, BHJ, MEH, VAP	Ahmedabad LF-3 → AHM-03

Then, per placemark:

- **Polygon with a year** → an outline of that site, part and epoch. Coordinates (longitude, latitude) are projected to UTM zone 43N (EPSG:32643); an invalid ring (self-crossing) is repaired with a zero-width buffer.
- **Pin with a year** → the site was checked in that epoch and that part had no waste (0). Two cases: Surat LF-2(a) 2020, Ahmedabad LF-2(b) 2026.
- **Pin without a year** → site location, used as the marker position in fig01 (only for pins of the whole site, not of a part).
- **Ignored:** Jamnagar LF-2 (remediated, not in the study) and any city not in the list. Every decision is written to the log (notes sheet of the Excel file).
- **A part with no placemark in an epoch counts as 0** once the site has any outline in that epoch. This gives Khajod (a) = 0 from 2020, Vadodara LF-2 old dump = 0 from 2022 and Vadodara LF-2 (a)/(b) = 0 in 2020.

Expected log lines (start of the console output):

```

IGNORED (Jamnagar LF-2 not in study): Jamnagar LF - 2 (remidiated)
Vapi-LF - 2025 -> VAP-01: image 2025 used as epoch 2024
Bhuj LF -2025 -> BHJ-01: image 2025 used as epoch 2024
Ahmedabad LF - 2(a) (2025) -> AHM-02: image 2025 used as epoch 2024
Ahmedabad Lf -2(b) (2026) (remediated) -> AHM-02 part b epoch 2026: no waste (0)
Ahmedabad LF-1 ( 2025) -> AHM-01: image 2025 used as epoch 2024
Ahmedabad LF -3 (2025) -> AHM-03: image 2025 used as epoch 2024
Surat LF - 2(a) 2020 Remediated -> SUR-02 part a epoch 2020: no waste (0)
OVERLAPS: none
sites: ['AHM-01', 'AHM-02', 'AHM-03', 'BHJ-01', 'BHV-01', 'GNR-01', 'JAM-01', 'MEH-01',
        'RAJ-01', 'SUR-01', 'SUR-02', 'VAD-01', 'VAD-02', 'VAP-01']

```

(Lines of the form "point (no epoch): ... -> SITE" list the 16 location pins and are omitted here.)

6. Step 1 - final_analysis.py (tables and data)

```
python final_analysis.py
```

Runs in about one second. What it does, in the order of the code:

6.1 Area, perimeter and uncertainty per site and epoch

1. For each part, all its polygons in that epoch are merged (union).
2. **Area** of the site = sum of the part areas in UTM 43N ($\text{m}^2 \div 10,000 = \text{ha}$). Parts are summed, not unioned (student decision); the script checks that parts do not overlap by more than 100 m^2 (none do).
3. **Perimeter** = sum of the outer-ring lengths of the parts after Douglas-Peucker simplification with 5 m tolerance (removes hand-tracing zig-zag; changes perimeter by at most 2.5% in the pilot). Holes are not counted.
4. **Edge uncertainty** ($\pm$) = $[\text{area}(\text{outline buffered } +2 \text{ m}) - \text{area}(\text{outline buffered } -2 \text{ m})] \div 2$, summed over parts (buffer method of Paul et al. for hand-digitised outlines). Drawn as the shaded band in fig03.
5. **Shape** from total area A and perimeter P: shape index = $P \div (2\sqrt{\pi A})$ (1 = circle); compactness = $4\pi A \div P^2$ (1 = circle, lower = more irregular or fragmented); fractal dimension = $2 \ln(P/4) \div \ln(A)$ (A in m^2 , P in m; FRAGSTATS form, about 1 for simple shapes, up to 2 for very convoluted edges).
6. **n_parts** = number of separate polygons after merging all parts of the epoch.

6.2 Trend of each site

- **Theil-Sen slope** (ha/yr): median of the slopes between every pair of epochs (15 pairs). Robust to one odd epoch.
- **Mann-Kendall S** = $\sum \text{sign}(\text{area}_j - \text{area}_i)$ over all pairs of epochs $i < j$; **tau** = $S \div 15$. **Exact p**: all 720 orderings of the six areas are enumerated and p = share of orderings with $|S| \geq$ the observed $|S|$ (two-sided). With $n = 6$ the smallest possible p is 0.003; p is descriptive only (no forecasting).
- Same Theil-Sen slope for perimeter (m/yr).

```
theil_sen = median( (y_j - y_i) / (x_j - x_i) for all i < j )
S          = sum( sign(y_j - y_i) for all i < j ), tau = S / 15
p_exact    = count( |S(permutation)| >= |S| ) / 720
```

6.3 Change descriptors

- Net change 2016 → 2026 (ha and %); largest extent and its year; change from largest extent to 2026 (%); smallest extent (trough), its year and change from trough to 2026.
- **Threshold** Th = $\max(15\%, 2 \times \text{mean relative edge uncertainty of the site})$. Th is 15% at 11 sites and larger at the three smallest or narrowest: Surat-Bhatar 19.7%, Vapi 21.1%, Vadodara LF-2 25.6%.
- **Significant reversals** = number of direction changes among epoch-to-epoch changes larger than Th (2 or more = volatile path).

6.4 Land turnover (overlay of outlines)

- **Ever under waste** = union of the outlines of all six epochs.
- **Newly covered since 2016** = 2026 outline minus 2016 outline (area of land under waste in 2026 that was not in 2016).
- **No longer waste in 2026** = ever under waste minus 2026 outline (land that held waste in at least one epoch but not in 2026: cleared, biominer, built over or reshaped).
- **Transitions** between consecutive epochs: gained, lost and persistent area, and overlap IoU = $\text{persistent} \div \text{union}$.

6.5 Trajectory class (first rule that applies)

Order	Class	Rule	Sites
1	Emerging	No waste in 2016, waste in 2026	-
2	Cleared	No waste in 2026	-
3	Cleared then regrowing	Net change < -Th, lowest area at an interior epoch, and 2026 more than Th above that low	SUR-02
4	Rise and fall	Largest area at an interior epoch, rise 2016 → largest > Th and fall largest → 2026 < -Th	AHM-02, BHV-01, MEH-01
5	Expanding	Net change > Th	AHM-03, BHJ-01, GNR-01, JAM-01, RAJ-01, VAD-01, VAP-01
6	Contracting	Net change < -Th	AHM-01, SUR-01, VAD-02
7	Stable	Otherwise	-

Because rule 4 comes before rule 5, Bhavnagar (+299% net) is "Rise and fall": it peaked in 2020 and fell 25% by 2026.

6.6 Totals and outputs

- City totals (sum of the city's sites) and group totals: "Major city" = Ahmedabad, Surat, Vadodara, Rajkot, Bhavnagar (9 sites); "Other city" = Gandhinagar, Jamnagar, Bhuj, Mehsana, Vapi (5 sites); "All sites".
- Excel workbook with six tables and a *notes* sheet (method notes + full processing log); the same tables as CSV.
- GeoJSON outlines back-projected to WGS84 (open in QGIS or Google Earth Pro).
- results.json: all rows, the summary, transitions, group and city totals, each outline as coordinates relative to the site centroid (UTM metres, simplified with 1 m tolerance, for drawing only) and the marker position of each site.

Expected console output ends with the per-site summary and the totals:

```
AHM-01  38.48 -> 29.93 net    -22% peak  38.7@2020 TS -1.01 tau -0.60 p 0.136 ... Contracting
...
VAP-01   0.53 -> 1.40 net   +167% peak   1.7@2022 TS +0.10 tau +0.60 p 0.136 ... Expanding
Major city {2016: 141.7, 2018: 156.1, 2020: 127.5, 2022: 146.2, 2024: 167.9, 2026: 134.8} -5%
All sites {2016: 152.5, 2018: 166.7, 2020: 142.7, 2022: 166.7, 2024: 192.1, 2026: 161.0} +6%
Other city {2016: 10.7, 2018: 10.6, 2020: 15.2, 2022: 20.5, 2024: 24.2, 2026: 26.2} +144%
Ahmedabad {2016: 50.0, ...} -7%    ... Vapi {2016: 0.5, ...} +167%
```

7. Step 2 - final_figures.py (figures)

```
python final_figures.py
```

Reads only data/results.json (and the state boundary for fig01) and prints one "saved ..." line per file: 26 lines (12 figures + 14 site maps). Takes a few seconds. Run Step 1 first whenever the KML changes.

7.1 Common style

- PNG at 200 dpi, white background, font DejaVu Sans, cropped tight to the content.
- Colours: five most populous cities = blue #2a78d6; other cities = orange #eb6834 (a colour-vision-deficiency-safe pair). Epochs in outline maps = a blue ramp from light (2016) to dark (2026), with the 2026 line thicker.
- Parts (fig10): old dump purple, (a) green, (b) pink, (c) amber, (a)+(b) traced together grey.
- Short site names on the figures: Ahmedabad-1 (Pirana), Ahmedabad-2, Ahmedabad-3, Surat-1 (Bhatar), Surat-2 (Khajod), Vadodara-1, Vadodara-2, Rajkot, Bhavnagar, Gandhinagar, Jamnagar, Bhuj, Mehsana, Vapi. Small-multiple figures order the sites blue group first, then orange, each by site ID.

7.2 Figure list

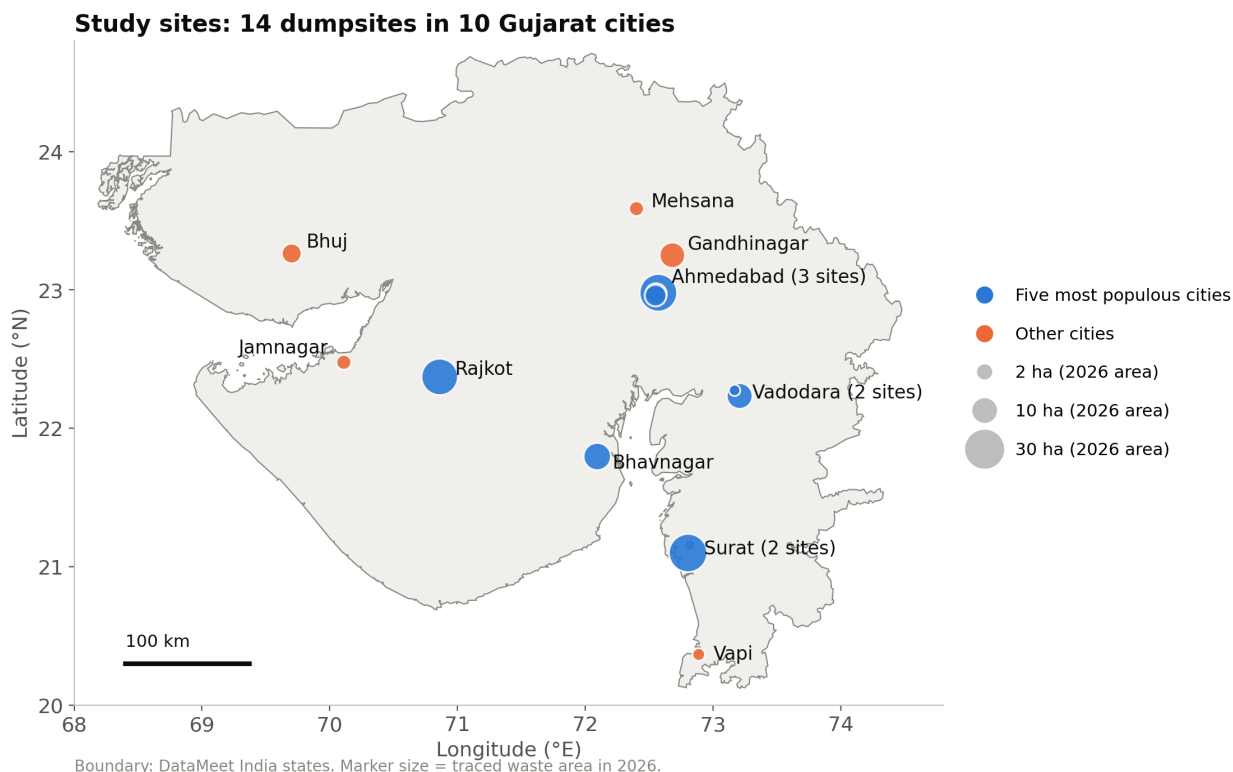
File	Figure type	Data from results.json
fig01_study_area_map.png	Map: state outline + one marker per site	centroids, summary (area_2026_ha, group)
fig02_total_footprint_by_group.png	Two line charts: hectares; index 2016 = 100	groups, rows (Khajod)
fig03_site_area_trends.png	14 small line charts with uncertainty band	rows (area_ha, area_unc_ha), summary (trajectory)
fig04_net_change_by_site.png	Horizontal bars	summary (net_change_ha, net_change_pct)
fig05_trajectory_quadrant.png	Scatter, log x-axis	summary (peak_ha, area_2016_ha, peak_to_2026_pct)
fig06_outlines_all_sites.png	14 outline maps	shapes, rows
fig08_land_turnover.png	Diverging horizontal bars	summary (newly_covered_since_2016_ha, no_longer_waste_2026_ha)
fig09_compactness_2016_2026.png	Dumbbell (2016 vs 2026)	summary (compactness_2016/2026, n_parts_max)
fig10_multipart_sites.png	Stacked bars per part	parts
fig11_site_perimeter_trends.png	14 small line charts	rows (perimeter_m), summary (perimeter_change_pct)
fig12_city_totals.png	Arrows 2016 → 2026 per city	cities
fig13_results_table.png	Table as image	summary
site_outlines/<SITE>_outlines.png	One outline map per site (14)	shapes, rows

There is no fig07: the number was left free when the figure set was finalised, and the script does not produce it.

8. What each figure shows

Each entry gives how the figure is built, how to read it, and the main numbers it carries (for your own wording).

fig01 - Study area map

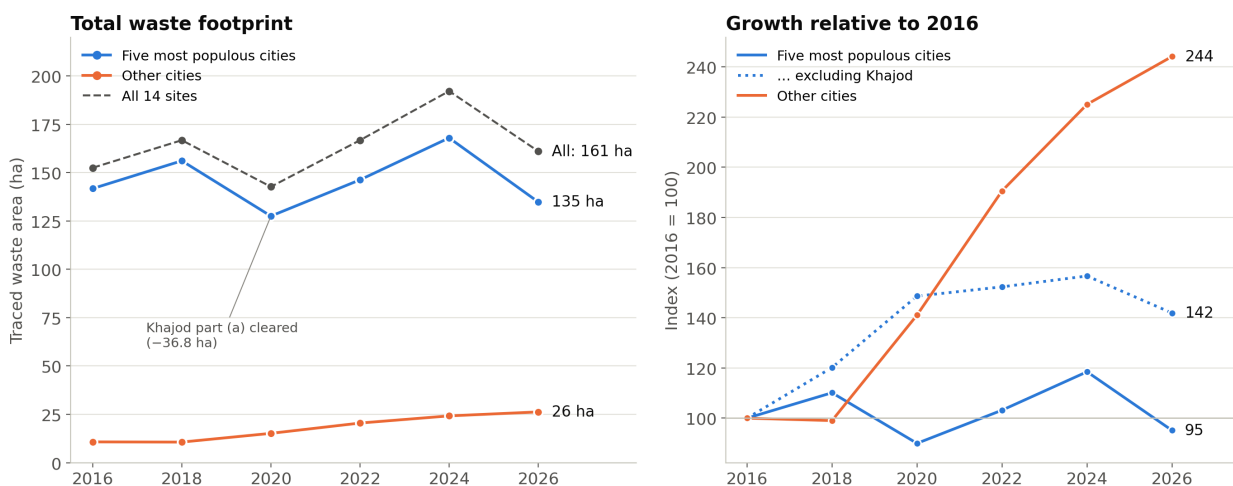


File: figures/fig01_study_area_map.png

How it is made. Gujarat outline (DataMeet) drawn in longitude/latitude with the aspect corrected for 22.5° N; one marker per site at its KML location pin (or the centre of all its outlines where the pin belongs to a part). Marker area grows with the 2026 waste area (legend: 2, 10, 30 ha). City labels are placed by hand; 100 km scale bar.

What it shows. Where the 14 sites are and how large each was in 2026. Blue = five most populous cities, orange = other cities. The largest 2026 markers are Surat-Khajod (31.2 ha), Pirana (29.9 ha) and Rajkot (28.1 ha).

fig02 - Total waste footprint by city group



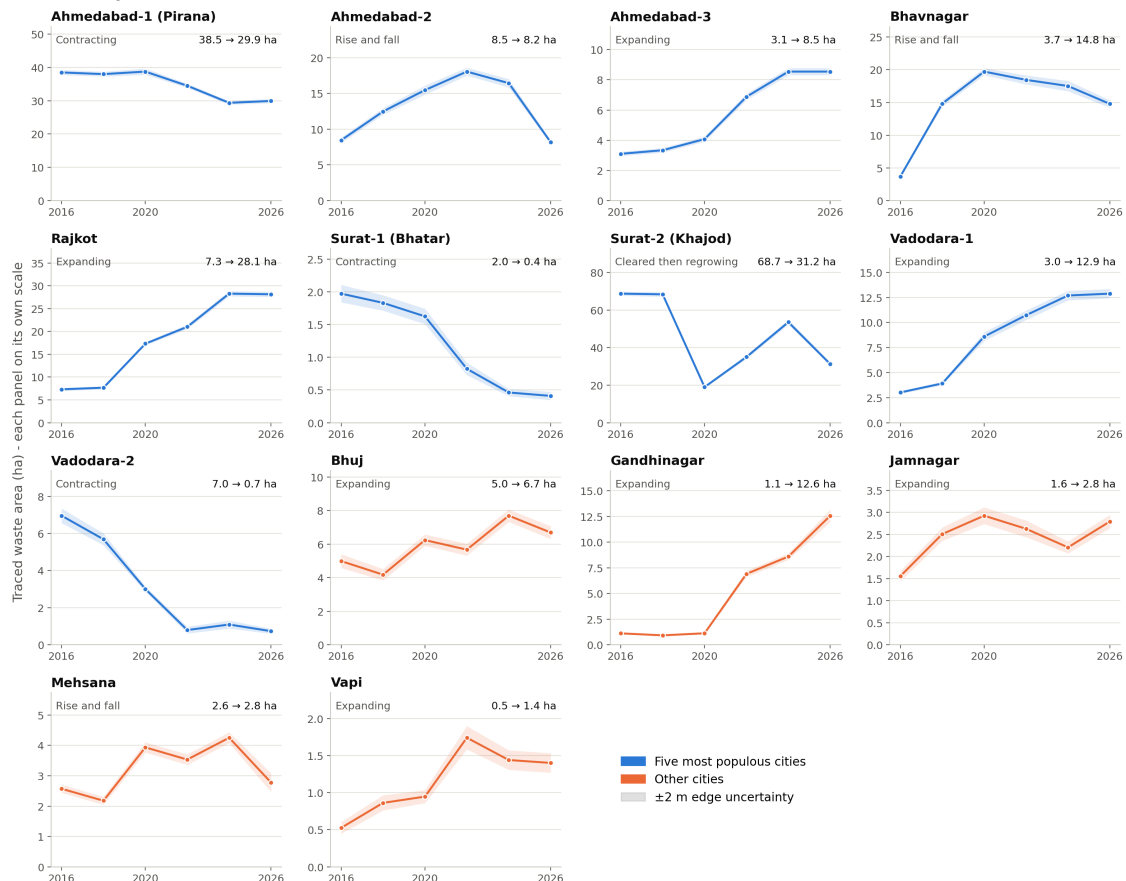
File: figures/fig02_total_footprint_by_group.png

How it is made. Left: sum of site areas per epoch for each group and for all 14 sites, with a note at the 2020 dip where Khajod part (a) (36.8 ha) was cleared. Right: the same series as an index (2016 = 100), plus the major-city series without Khajod (dotted).

What it shows. All sites 152.5 → 192.1 (2024) → 161.0 ha. Major cities end at index 95, but 142 without Khajod; other cities rise to 244 and are above 2016 in every epoch after 2018. One large clearance hides growth elsewhere.

fig03 - Waste area per site

Waste area per site, 2016-2026

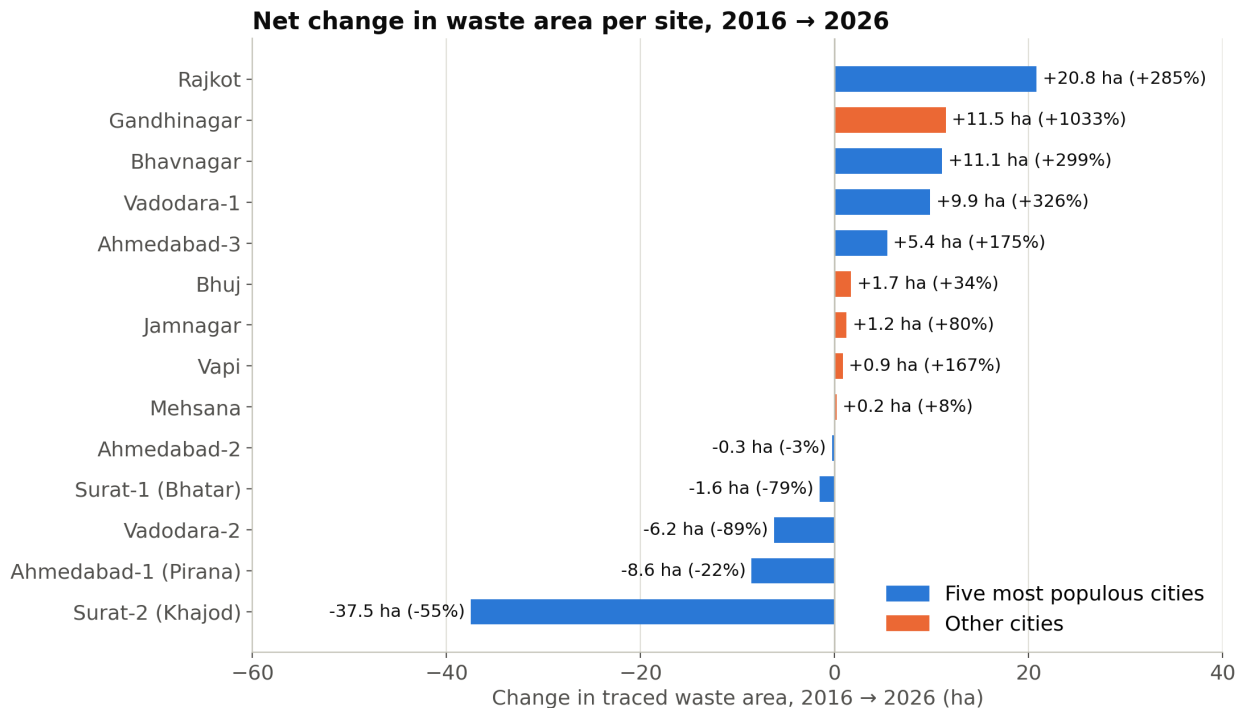


File: figures/fig03_site_area_trends.png

How it is made. One panel per site, each with its own y-axis starting at 0. Line = traced area; shaded band = ±2 m edge uncertainty; top-left = trajectory class; top-right = 2016 → 2026 area.

What it shows. Shape of each site's path, e.g. Khajod's drop in 2020 and regrowth to 2024, Gandhinagar's jump after 2020, Pirana's decline after 2020. Because scales differ, compare shapes, not heights, between panels.

fig04 - Net change per site

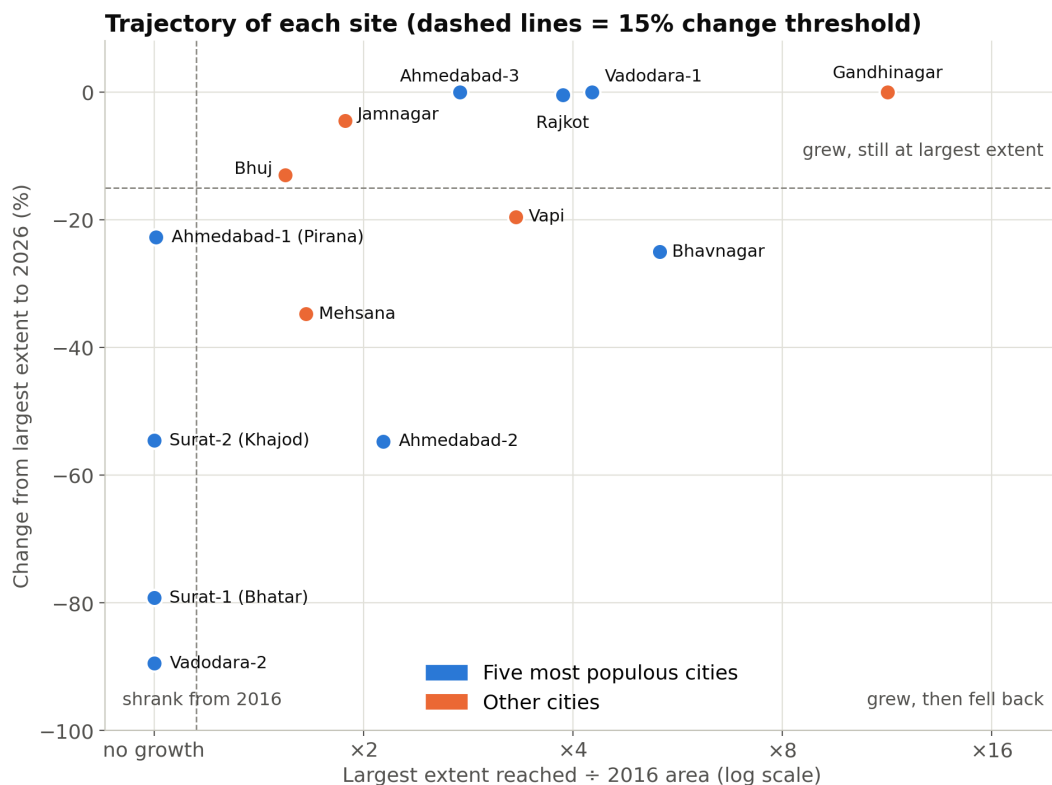


File: figures/fig04_net_change_by_site.png

How it is made. Horizontal bars of 2026 area minus 2016 area, sorted; labels give hectares and percent.

What it shows. Rajkot +20.8 ha is the largest gain; Khajod -37.5 ha the largest loss. Percentages are large where the 2016 area was small (Gandhinagar +1033% from 1.1 ha).

fig05 - Trajectory quadrant



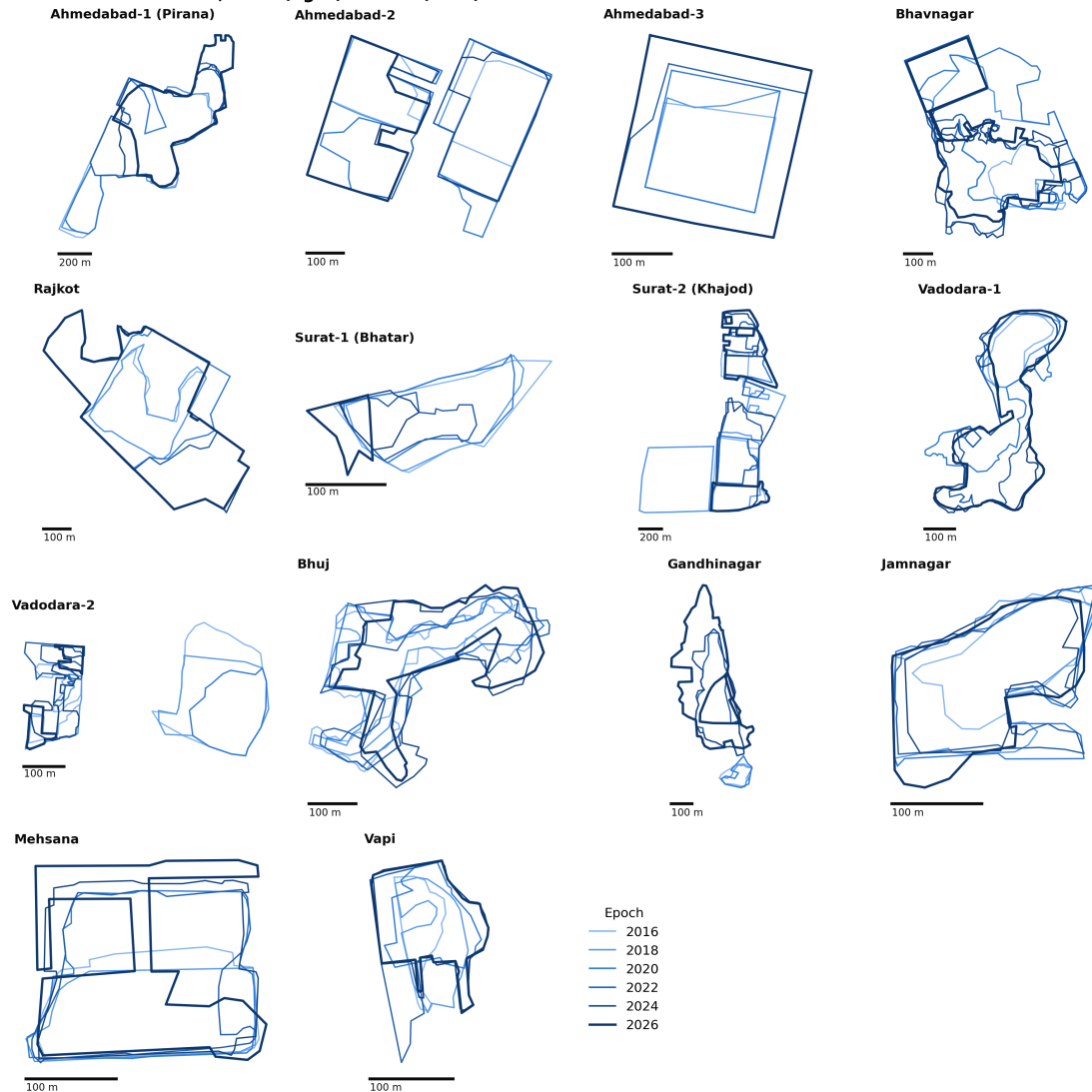
File: figures/fig05_trajectory_quadrant.png

How it is made. $x = \text{largest extent reached} \div 2016 \text{ area (log scale, "no growth" = 1)}$; $y = \text{change from the largest extent to 2026 (\%)}$. Dashed lines at $\times 1.15$ and -15% mark the 15% change threshold.

What it shows. Top right (above -15%): grew and still at or near the largest extent (Gandhinagar, Vadodara-1, Rajkot, Ahmedabad-3; Jamnagar and Bhuj less than 15% below it). Lower right: grew, then fell back more than 15% (Ahmedabad-2, Bhavnagar, Mehsana, Vapi). Left edge: largest in 2016, then shrank (Vadodara-2, Surat-1, Khajod, and Pirana, which peaked at 2020 only 0.7% above 2016).

fig06 - Outlines of all sites

Traced waste outlines, 2016 (light) → 2026 (dark)

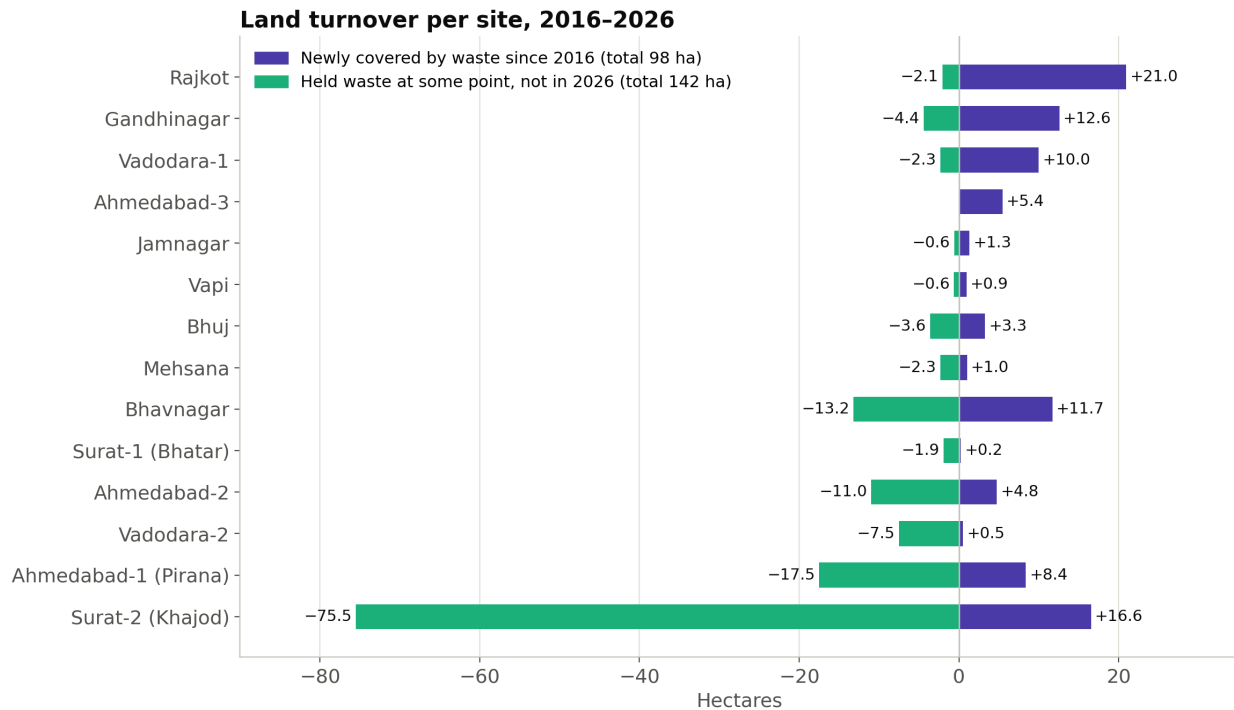


File: figures/fig06_outlines_all_sites.png

How it is made. Every traced outline, one panel per site, epochs from light (2016) to dark, thick (2026). Each panel has its own scale bar (100, 200 or 500 m chosen from the site size). Outlines are simplified with 1 m tolerance for drawing only.

What it shows. How each footprint moved: e.g. Khajod's south-west block (part a) disappears after 2018; Gandhinagar spreads north; Vadodara-2's eastern old dump disappears.

fig08 - Land turnover

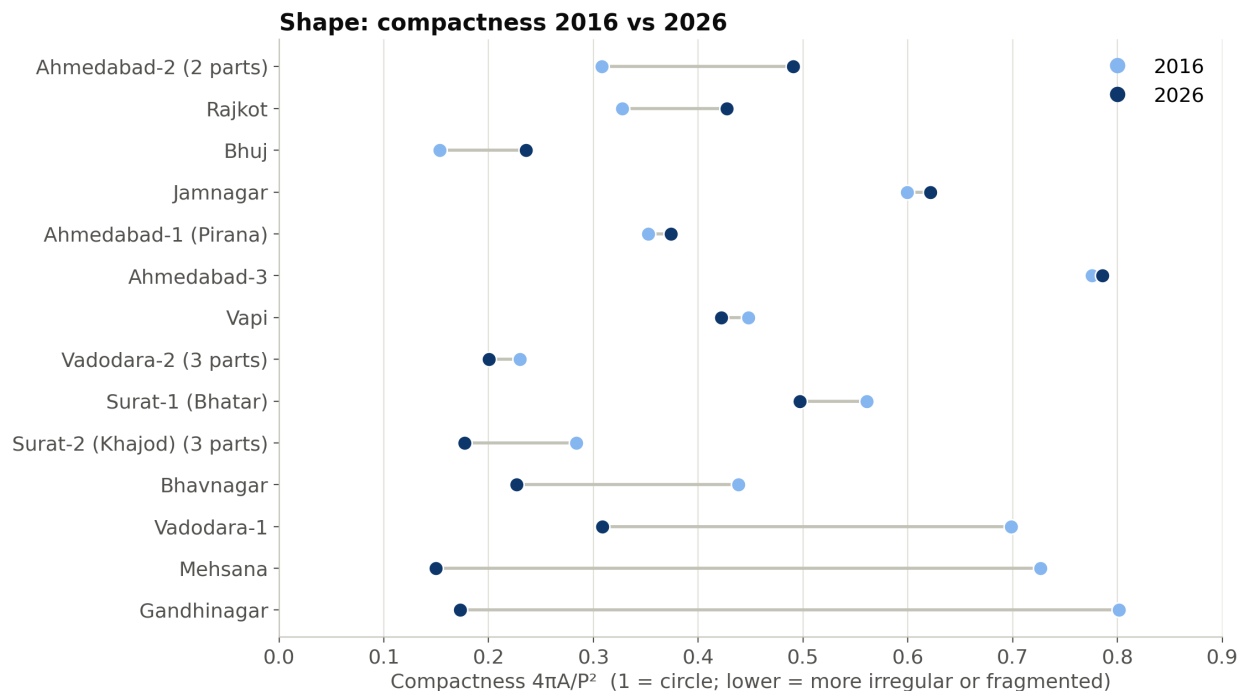


File: figures/fig08_land_turnover.png

How it is made. Right (purple): land newly covered by waste since 2016 (2026 outline minus 2016 outline). Left (green): land that held waste at some epoch but not in 2026 (union of all outlines minus the 2026 outline). Totals in the legend.

What it shows. 97.7 ha newly covered vs 142.4 ha no longer under waste. Khajod dominates the green side (−75.5 ha); Rajkot, Gandhinagar and Khajod dominate the purple side.

fig09 - Compactness 2016 vs 2026



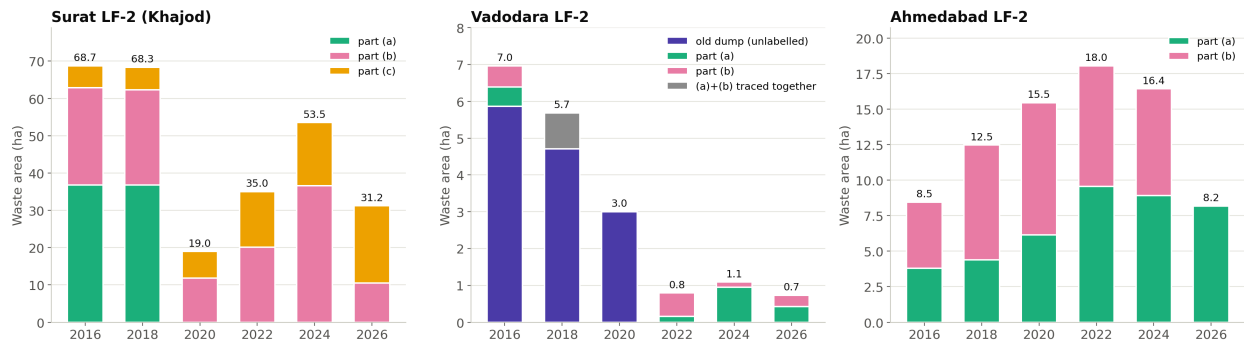
File: figures/fig09_compactness_2016_2026.png

How it is made. Dumbbell per site: light dot 2016, dark dot 2026, compactness $4\pi A/P^2$ (1 = circle). Sorted by change; multi-part sites are labelled with their number of parts.

What it shows. Large drops = footprints becoming irregular or fragmented (Gandhinagar 0.80 → 0.17, Mehsana 0.73 → 0.15, Vadodara-1 0.70 → 0.31). Rises = more regular footprints (Ahmedabad-2 0.31 → 0.49, Rajkot 0.33 → 0.43, Bhuj 0.15 → 0.24).

fig10 - Sites traced in parts

Sites traced in parts: area of each part (stacked = site total)



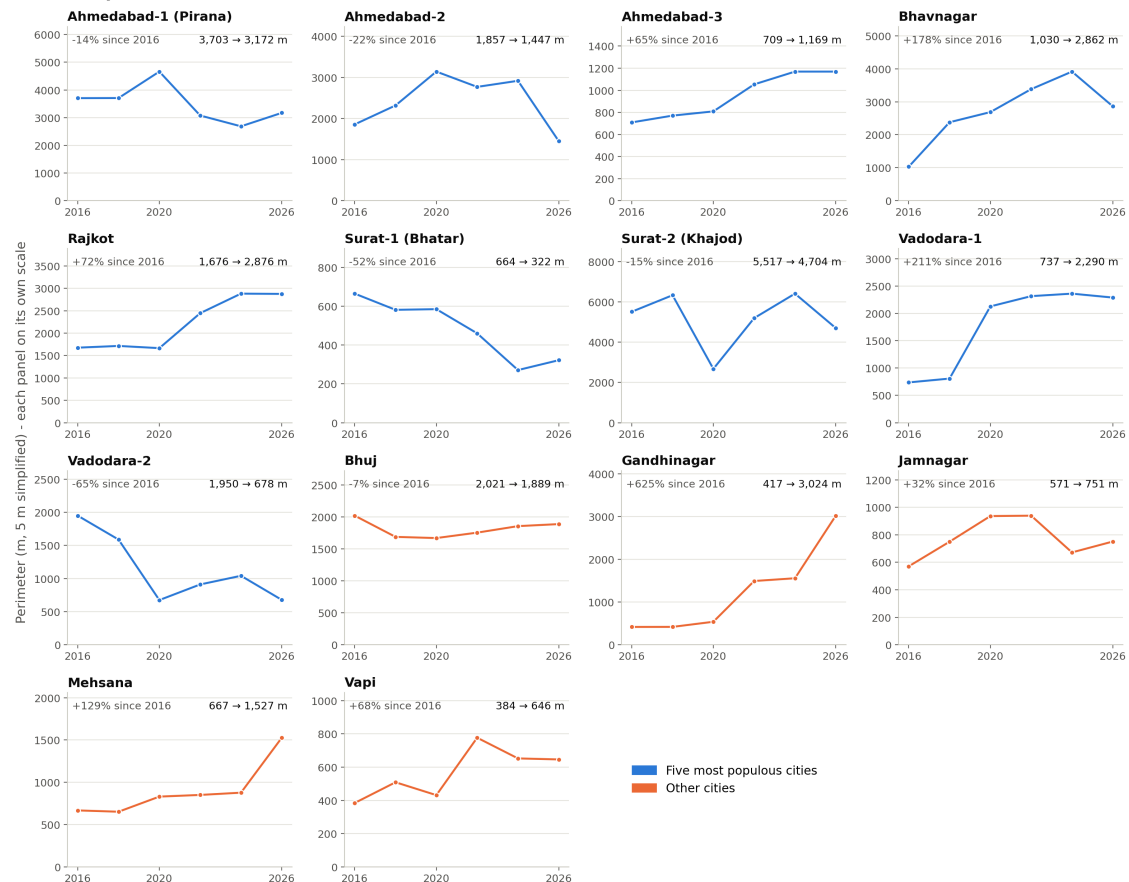
File: figures/fig10_multipart_sites.png

How it is made. Stacked bars of each part's area per epoch for Khajod, Vadodara LF-2 and Ahmedabad LF-2; the label on top is the site total used everywhere else.

What it shows. Khajod: part (a) 36.8 ha cleared by 2020; (b) and (c) regrow. Vadodara LF-2: old dump gone by 2022; small (a)/(b) patches after 2020. Ahmedabad LF-2: part (b) remediated by 2026.

fig11 - Perimeter per site

Perimeter per site, 2016-2026

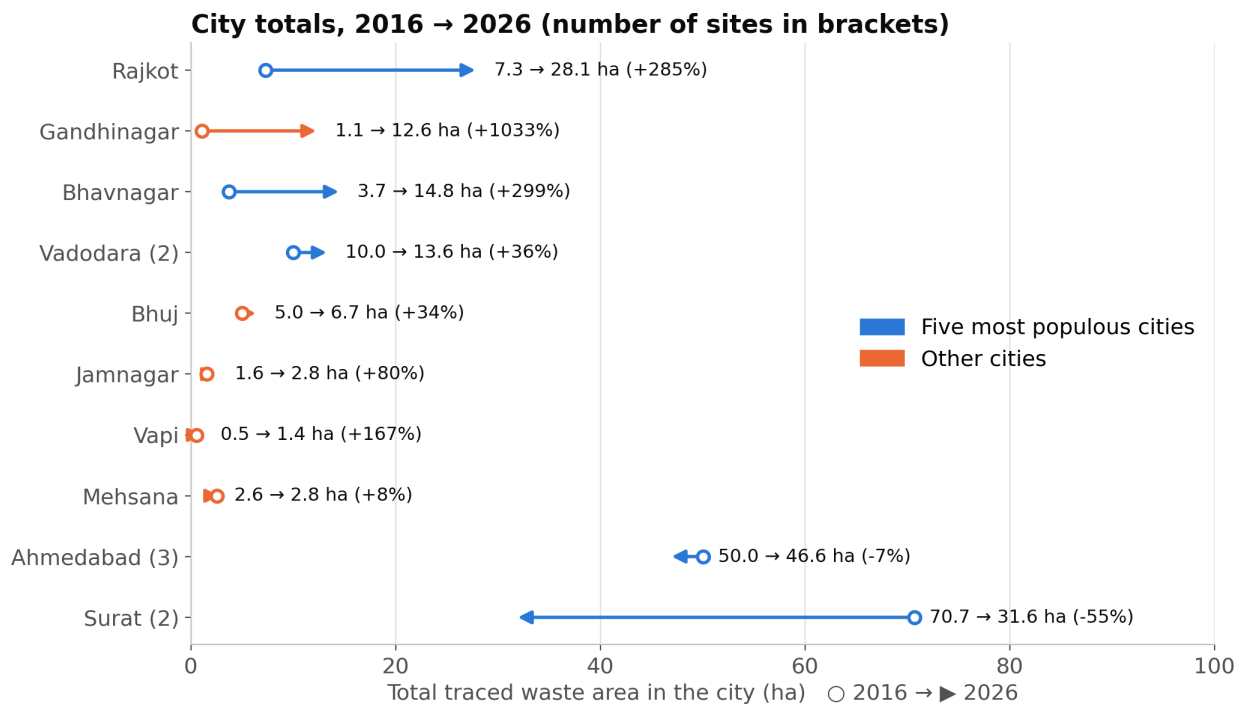


File: figures/fig11_site_perimeter_trends.png

How it is made. As fig03 but for perimeter (5 m simplified, parts summed); top-left = perimeter change since 2016.

What it shows. Perimeter tracks both size and edge complexity: Gandhinagar 417 → 3,024 m, Bhavnagar 1,030 → 2,862 m, Vadodara-2 1,950 → 678 m.

fig12 - City totals



File: figures/fig12_city_totals.png

How it is made. Per city: open circle = 2016 total, arrow head = 2026 total; label gives both and the percent change. Number of sites in brackets.

What it shows. Surat 70.7 → 31.6 ha (–55%), Ahmedabad 50.0 → 46.6 (–7%); growth in Rajkot (+285%), Bhavnagar (+299%), Gandhinagar (+1033%), Vadodara (+36%). For Vapi and Mehsana the change is smaller than the arrow head, so the head is drawn over the circle (see Section 12).

fig13 - Results table

Results per site (Theil-Sen trend; Mann-Kendall exact p, n = 6; blue = five most populous cities)

Site	2016 ha	2026 ha	Net %	Largest (year)	Trend ha/yr	MK p	Perimeter (m)	Class
Ahmedabad-1 (Pirana)	38.5	29.9	-22%	38.7 (2020)	-1.01	0.136	3,703 → 3,172	Contracting
Ahmedabad-2	8.5	8.2	-3%	18.0 (2022)	+0.66	0.719	1,857 → 1,447	Rise and fall
Ahmedabad-3	3.1	8.5	+175%	8.5 (2024)	+0.65	0.006	709 → 1,169	Expanding
Bhavnagar	3.7	14.8	+299%	19.7 (2020)	+0.45	1.000	1,030 → 2,862	Rise and fall
Rajkot	7.3	28.1	+285%	28.3 (2024)	+2.50	0.017	1,676 → 2,876	Expanding
Surat-1 (Bhatar)	2.0	0.4	-79%	2.0 (2016)	-0.18	0.003	664 → 322	Contracting
Surat-2 (Khajod)	68.7	31.2	-55%	68.7 (2016)	-2.47	0.272	5,517 → 4,704	Cleared then regrowing
Vadodara-1	3.0	12.9	+326%	12.9 (2026)	+1.07	0.003	737 → 2,290	Expanding
Vadodara-2	7.0	0.7	-89%	7.0 (2016)	-0.64	0.017	1,950 → 678	Contracting
Bhuj	5.0	6.7	+34%	7.7 (2024)	+0.31	0.136	2,021 → 1,889	Expanding
Gandhinagar	1.1	12.6	+1033%	12.6 (2026)	+1.28	0.017	417 → 3,024	Expanding
Jamnagar	1.6	2.8	+80%	2.9 (2020)	+0.04	0.469	571 → 751	Expanding
Mehsana	2.6	2.8	+8%	4.3 (2024)	+0.08	0.469	667 → 1,527	Rise and fall
Vapi	0.5	1.4	+167%	1.7 (2022)	+0.10	0.136	384 → 646	Expanding

File: figures/fig13_results_table.png

How it is made. Summary per site: 2016 and 2026 area, net %, largest extent (year), Theil-Sen trend, Mann-Kendall exact p, perimeter 2016 → 2026, class. Site names coloured by group.

What it shows. The one-page reference for all per-site numbers; Section 11 has the same values with more columns.

Site outline maps (figures/site_outlines/)

One larger map per site, drawn exactly like fig06 but with the traced area of each epoch in the legend. All 14 are in Appendix A. Use them for case-study slides (e.g. Khajod, Pirana, Rajkot, Vadodara LF-2).

9. Manual check of one site (Google Earth Pro and QGIS)

This recomputes the numbers of one outline by hand, without the scripts. Example: Bhavnagar LF-1, 2026 (placemark Bhavnagar LF -2026).

9.1 Google Earth Pro (quick check)

1. Open the master KML in Google Earth Pro. In *Places*, open the Bhavnagar folder, right-click Bhavnagar LF -2026 ► Properties ► *Measurements* tab.
2. Expected: area $\approx 147,700 \text{ m}^2$ (14.77 ha) and perimeter $\approx 2,868 \text{ m}$. Google Earth measures on the ellipsoid; the script works in UTM 43N and gives 14.79 ha and 2,862 m (after 5 m simplification; 2,870 m before). The 0.14% area difference comes from the UTM projection, which enlarges areas slightly: from 0.003% at Vadodara to 0.65% at Bhuj, the site furthest from the zone centre at 75° E.

9.2 QGIS (exact check)

1. Layer ► Add Layer ► Add Vector Layer ► the master KML. If QGIS asks which layers to add, pick *Bhavnagar*.
2. Select the outline: Select Features by Expression, "Name" = 'Bhavnagar LF -2026'. Right-click the layer ► Export ► Save Selected Features As..., format GeoPackage, CRS **EPSG:32643 - WGS 84 / UTM zone 43N**.
3. Open the attribute table of the new layer ► Field Calculator, virtual field, expression $\text{area}(@\text{geometry}) / 10000$ (planar area in the layer CRS). Expected **14.7888** ha = area_ha of BHV-01 2026 in tables/site_by_epoch.csv.
4. Perimeter: $\text{perimeter}(@\text{geometry}) \rightarrow 2,869.8 \text{ m}$ (raw). Then Vector geometry ► Simplify, Douglas-Peucker, tolerance 5 m, and the same expression $\rightarrow$ **2,862.2** m = perimeter_m.
5. Uncertainty: Vector geometry ► Buffer with distance 2 and with -2 . Areas 15.3607 and 14.2180 ha $\rightarrow (15.3607 - 14.2180) \div 2 =$ **0.571** ha = area_unc_ha (3.9%).
6. Compactness: $4\pi \times 147,888 \div 2,862.2^2 =$ **0.227** = compactness_2026 of BHV-01 in tables/site_summary.csv.

A second site to try: Jamnagar LF-1 2016 $\rightarrow$ area 1.5549 ha, perimeter 576.3 m raw / 570.9 m simplified, buffer areas 1.6712 and 1.4412 ha (uncertainty 0.115 ha).

9.3 Checking a trend by hand

Take the six areas of one site from tables/site_by_epoch.csv (e.g. Vadodara LF-1: 3.02, 3.92, 8.59, 10.72, 12.69, 12.88 ha). All 15 later-minus-earlier differences are positive, so $S = +15$, $\tau = 1.0$ and exact $p = 2 \div 720 = 0.003$ (only the fully increasing and fully decreasing orders reach $|S| = 15$). The Theil-Sen slope is the median of the 15 pairwise slopes = +1.07 ha/yr, as in fig13.

10. Checks carried out

Check	How	Result
Reproducibility of tables	Fresh copy of the repository; ran Step 1; compared the six CSVs with the committed ones	6 of 6 byte-identical
Reproducibility of data files	results.json and both GeoJSONs compared value by value	Identical to 1 part in a million (last-digit float formatting only)
Reproducibility of figures	Ran Step 2; compared the 26 PNGs with the committed ones	26 of 26 produced; same content, image sizes differ by 1-11 pixels (matplotlib version / font rendering)
Completeness	Status of every site-epoch	84 of 84 (14 × 6) traced; none missing
Parts do not overlap	Sum of parts vs union of parts, every site-epoch	No overlap > 100 m ² ("OVERLAPS: none")
Projection	Geodesic area on the WGS84 ellipsoid (pyproj.Geod) vs UTM 43N area, 84 site-epochs	UTM 0.003-0.65% larger (Vadodara smallest, Bhuj largest: distance from 75° E)

Check	How	Result
Perimeter simplification	UTM perimeter after 5 m simplification vs raw geodesic perimeter	Median −0.2%, range −2.8% to +0.1%
Theil-Sen	Script vs scipy.stats.theilslopes, 14 sites	Agree (max difference 0.00005 ha/yr = rounding)
Mann-Kendall tau	Script vs scipy.stats.kendalltau	Agree except AHM-03 (0.933 vs 0.966): the script uses tau-a ($S \div 15$) and AHM-03 has a tie (2024 = 2026); S and p are unaffected
Totals	Major + other = all sites, every epoch; sum of city totals = all sites	Exact
Transitions	gained – lost = change in area, 70 epoch pairs	Exact (≤ 0.0001 ha)
Manual	QGIS / Google Earth Pro, Section 9	Area, perimeter, uncertainty and compactness reproduced
Visual	All 26 figures inspected	OK; quirks listed in Section 12

11. Full results

11.1 Area per site and epoch (ha)

ID	Site	2016	2018	2020	2022	2024	2026	Mean ±
AHM-01	Ahmedabad LF-1 (Pirana)	38.48	37.98	38.73	34.49	29.36	29.93	2.0%
AHM-02	Ahmedabad LF-2	8.45	12.47	15.46	18.04	16.43	8.18	3.7%
AHM-03	Ahmedabad LF-3	3.10	3.34	4.08	6.88	8.55	8.55	3.6%
BHV-01	Bhavnagar LF-1	3.70	14.83	19.72	18.46	17.52	14.79	3.9%
RAJ-01	Rajkot LF-1	7.31	7.67	17.31	21.01	28.28	28.14	2.9%
SUR-01	Surat LF-1 (Bhatar)	1.97	1.83	1.62	0.82	0.46	0.41	9.8%
SUR-02	Surat LF-2 (Khajod)	68.72	68.35	18.98	35.01	53.54	31.21	2.4%
VAD-01	Vadodara LF-1	3.02	3.92	8.59	10.72	12.69	12.88	4.3%
VAD-02	Vadodara LF-2	6.96	5.69	3.01	0.79	1.09	0.73	12.8%
BHJ-01	Bhuj LF-1	4.99	4.18	6.24	5.67	7.69	6.69	6.4%
GNR-01	Gandhinagar LF-1	1.11	0.92	1.12	6.91	8.60	12.58	6.5%
JAM-01	Jamnagar LF-1	1.55	2.51	2.92	2.63	2.21	2.79	6.4%
MEH-01	Mehsana LF-1	2.57	2.18	3.94	3.53	4.25	2.78	5.9%
VAP-01	Vapi LF-1	0.53	0.86	0.95	1.74	1.44	1.40	10.6%
	Five most populous cities	141.7	156.1	127.5	146.2	167.9	134.8	
	Other cities	10.7	10.6	15.2	20.5	24.2	26.2	
	All 14 sites	152.5	166.7	142.7	166.7	192.1	161.0	

Mean ± = mean edge uncertainty of the site as a percentage of its area (±2 m buffer).

11.2 Trend, shape and turnover per site

ID	Theil-Se n (ha/yr)	MK S	MK p	Perimeter 2016 → 2026 (m)	Compact. 2016 → 2026	Newly covered (ha)	No longer waste (ha)	Th	Class
AHM-01	-1.01	-9	0.136	3,703 → 3,172	0.35 → 0.37	8.4	17.5	15%	Contracting
AHM-02	+0.66	+3	0.719	1,857 → 1,447	0.31 → 0.49	4.8	11.0	15%	Rise and fall
AHM-03	+0.65	+14	0.006	709 → 1,169	0.78 → 0.79	5.4	0.0	15%	Expanding
BHV-01	+0.45	+1	1.000	1,030 → 2,862	0.44 → 0.23	11.7	13.2	15%	Rise and fall
RAJ-01	+2.50	+13	0.017	1,676 → 2,876	0.33 → 0.43	21.0	2.1	15%	Expanding
SUR-01	-0.18	-15	0.003	664 → 322	0.56 → 0.50	0.2	1.9	20%	Contracting
SUR-02	-2.47	-7	0.272	5,517 → 4,704	0.28 → 0.18	16.6	75.5	15%	Cleared then regrowing
VAD-01	+1.07	+15	0.003	737 → 2,290	0.70 → 0.31	10.0	2.3	15%	Expanding
VAD-02	-0.64	-13	0.017	1,950 → 678	0.23 → 0.20	0.5	7.5	26%	Contracting
BHJ-01	+0.31	+9	0.136	2,021 → 1,889	0.15 → 0.24	3.3	3.6	15%	Expanding
GNR-01	+1.28	+13	0.017	417 → 3,024	0.80 → 0.17	12.6	4.4	15%	Expanding
JAM-01	+0.04	+5	0.469	571 → 751	0.60 → 0.62	1.3	0.6	15%	Expanding
MEH-01	+0.08	+5	0.469	667 → 1,527	0.73 → 0.15	1.0	2.3	15%	Rise and fall
VAP-01	+0.10	+9	0.136	384 → 646	0.45 → 0.42	0.9	0.6	21%	Expanding

11.3 City totals (ha)

City	Sites	2016	2018	2020	2022	2024	2026	Net
Surat	2	70.7	70.2	20.6	35.8	54.0	31.6	-55%
Ahmedabad	3	50.0	53.8	58.3	59.4	54.3	46.6	-7%
Mehsana	1	2.6	2.2	3.9	3.5	4.3	2.8	+8%
Vapi	1	0.5	0.9	0.9	1.7	1.4	1.4	+167%

City	Sites	2016	2018	2020	2022	2024	2026	Net
Jamnagar	1	1.6	2.5	2.9	2.6	2.2	2.8	+80%
Bhuj	1	5.0	4.2	6.2	5.7	7.7	6.7	+34%
Vadodara	2	10.0	9.6	11.6	11.5	13.8	13.6	+36%
Bhavnagar	1	3.7	14.8	19.7	18.5	17.5	14.8	+299%
Gandhinagar	1	1.1	0.9	1.1	6.9	8.6	12.6	+1033%
Rajkot	1	7.3	7.7	17.3	21.0	28.3	28.1	+285%

12. Notes and quirks found during the check

None of these changes a number in the tables. They are worth knowing before presenting.

Item	What	Suggested handling
Same outline in two epochs	Two outlines are exact copies of the previous epoch: Ahmedabad LF-3 2024 = 2026 (8.55 ha) and Khajod part (a) 2016 = 2018 (36.81 ha).	Fine if the imagery showed no visible change; confirm in Google Earth Pro that this was intended and not a copy left unedited.
fig05 dashed lines	Drawn at 15% for all sites, but the class threshold is higher at Surat-Bhatar (19.7%), Vapi (21.1%) and Vadodara LF-2 (25.6%).	Say "15% (higher for three small sites)" if you describe the lines.
fig12 arrow heads	For Vapi (+0.9 ha) and Mehsana (+0.2 ha) the change is shorter than the arrow head, so the head is drawn over the circle and can look like a decrease.	Read the label (+167%, +8%); or say so if asked.
No fig07	Numbering jumps from fig06 to fig08.	Renumber on the slides if needed.
fig02 note	"Khajod part (a) cleared (−36.8 ha)" refers to the 2018 → 2020 change of part (a) only; the whole site fell by 49.4 ha that step (68.3 → 19.0 ha).	Keep the two numbers apart.
Mann-Kendall tau	tau-a ($S \div 15$) is reported; with the tie at Ahmedabad LF-3 tau-b would be 0.966 instead of 0.933.	Quote S and p, which are exact.
Rajkot (open point)	Traced area grows 7.3 → 28.1 ha while news reports describe the old Nakrawadi dump turned into an urban forest. The volume guide found fill of up to ~14 m inside the traced footprint by Jan 2018.	Confirm the traced polygon is the active dumping area beside the remediated part.
Results web page	gujarat_dumpsite_footprints_final.html is not rebuilt by the scripts.	If the KML changes, re-run Steps 1 and 2 and treat the page as outdated.

13. Limitations

- One analyst traced all outlines; the planned blind re-trace (repeatability) was pending at the time of the pilot. Edge uncertainty (± 2 m buffer) covers edge placement, not interpretation.
- Imagery dates differ within an epoch (season, month) and a 2025 image stands in for 2024 at five sites.
- Area is not volume: a site can shrink on the map while its mound grows higher (see the volume guide), and waste below ground is invisible.
- Six epochs: Mann-Kendall p is descriptive only (smallest possible $p = 0.003$) and no forecasting is done.
- Class boundaries depend on the threshold (15% or $2 \times$ edge uncertainty); a site near the threshold can change class with a small tracing change.
- Perimeter depends on tracing detail; it is reported after 5 m simplification to reduce this. Holes inside outlines are not counted in the perimeter.
- Land turnover uses only the 2016 and 2026 outlines for "newly covered"; land that became waste and was cleared between those years counts only in "no longer waste".
- The five-most-populous vs other-cities grouping is by city, not by site size or age.

14. Troubleshooting

Message	Cause and fix
ModuleNotFoundError: No module named ...	Run <code>python -m pip install -r requirements.txt</code> in the scripts folder.
FileNotFoundError: ... completer.kml	The KML must be in the repository root with exactly the name "Locations of landfills in gujurat completer.kml".
FileNotFoundError: ... results.json (Step 2)	Run Step 1 first.
PermissionError on final_results.xlsx	The workbook is open in Excel. Close it and re-run Step 1.

Message	Cause and fix
KeyError: 'XXX-01' (e.g. a new site)	A new site ID is not in the fixed lists NAMES (Step 1) and SHORT (Step 2); add it to both (Appendix C).
A new outline does not appear	Check the notes sheet / console log for "IGNORED" and the name rules in Section 5.2 (city first, LF number, part in brackets, year).
A site-epoch has status "missing"	No placemark for that site in that epoch. Add the outline, or a pin named with the year and "remediated" if there was no waste.
Figure sizes differ by a few pixels	Different matplotlib version; content is the same (Section 10).

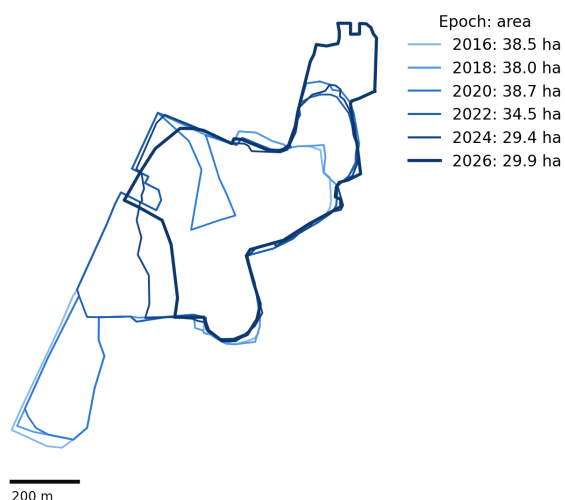
15. References and links

- Paul, F. et al. (2013). On the accuracy of glacier outlines derived from remote-sensing data. *Annals of Glaciology* 54(63): 171-182. ($\pm$ buffer method for the uncertainty of hand-digitised outlines; verify before citing.)
- Sen, P. K. (1968). Estimates of the regression coefficient based on Kendall's tau. *Journal of the American Statistical Association* 63: 1379-1389; Theil, H. (1950). A rank-invariant method of linear and polynomial regression analysis. *Proc. KNAW* 53. (Theil-Sen slope.)
- Mann, H. B. (1945). Nonparametric tests against trend. *Econometrica* 13: 245-259; Kendall, M. G. (1975). *Rank Correlation Methods*. (Mann-Kendall test.)
- McGarigal, K. & Marks, B. J. (1995). FRAGSTATS: spatial pattern analysis program for quantifying landscape structure. USDA Forest Service GTR PNW-351. (Shape index, fractal dimension; verify before citing.)
- Douglas, D. H. & Peucker, T. K. (1973). Algorithms for the reduction of the number of points required to represent a digitized line or its caricature. *Cartographica* 10(2): 112-122. (Perimeter simplification.)
- Data: Google Earth Pro (historical imagery) · DataMeet Community Maps, github.com/datameet/maps (state boundary) · EPSG:32643 WGS 84 / UTM zone 43N.
- Software: Python with shapely, pyproj, pandas, numpy, matplotlib, openpyxl · QGIS.
- Other project notes: README.md (figure list, key numbers), context.md (project log, tracing and pilot validation), volume_final_26Sep guide (height and volume).

Appendix A. Outline map of every site

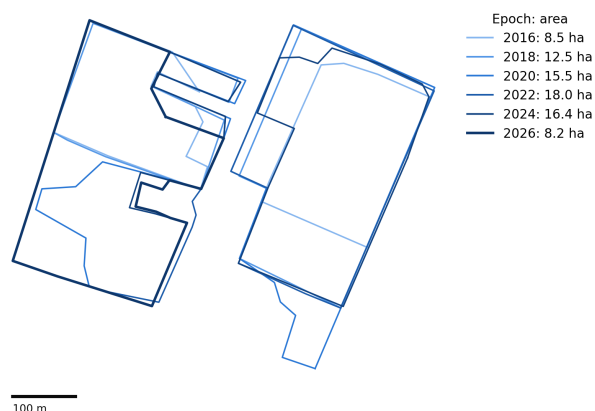
Traced outlines per epoch from light (2016) to dark (2026); legend = traced area per epoch. Same drawing as fig06, larger.

Ahmedabad LF-1 (Pirana): traced outlines



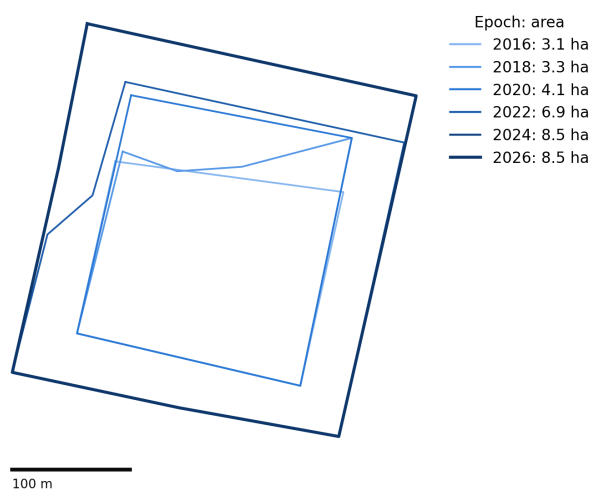
AHM-01 Ahmedabad LF-1 (Pirana): 38.5 → 29.9 ha. Contracting. Largest in 2020 (38.7 ha); −22% by 2026 while biomining.

Ahmedabad LF-2: traced outlines



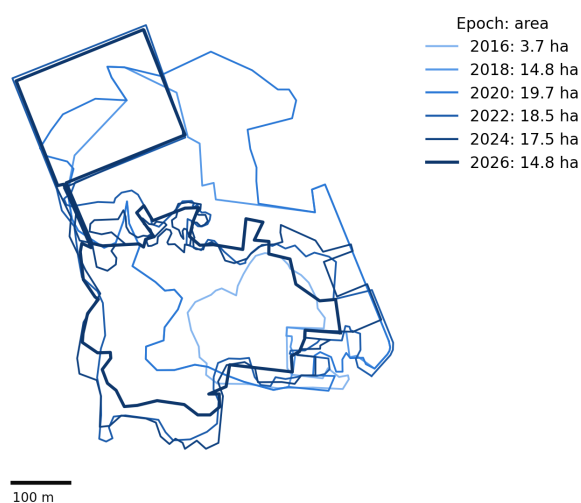
AHM-02 Ahmedabad LF-2: 8.5 → 8.2 ha. Rise and fall. Two parts ~300 m apart; part (b) remediated by 2026.

Ahmedabad LF-3: traced outlines



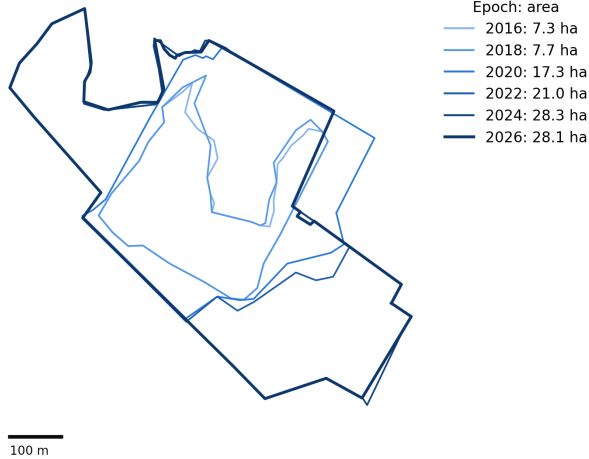
AHM-03 Ahmedabad LF-3: 3.1 → 8.5 ha. Expanding. Walled cell; 2024 and 2026 outlines identical.

Bhavnagar LF-1: traced outlines



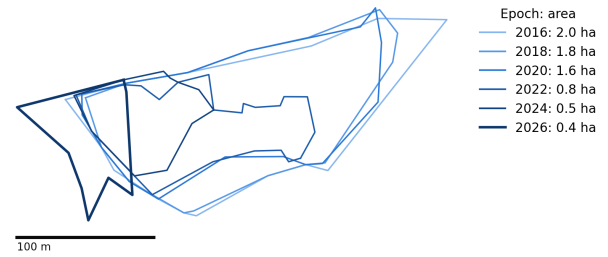
BHV-01 Bhavnagar LF-1: 3.7 → 14.8 ha. Rise and fall. 3.7 ha → 19.7 ha in 2020 → 14.8 ha.

Rajkot LF-1: traced outlines



RAJ-01 Rajkot LF-1: 7.3 → 28.1 ha. Expanding. 7.3 → 28.1 ha (open point, Section 12).

Surat LF-1 (Bhatar): traced outlines



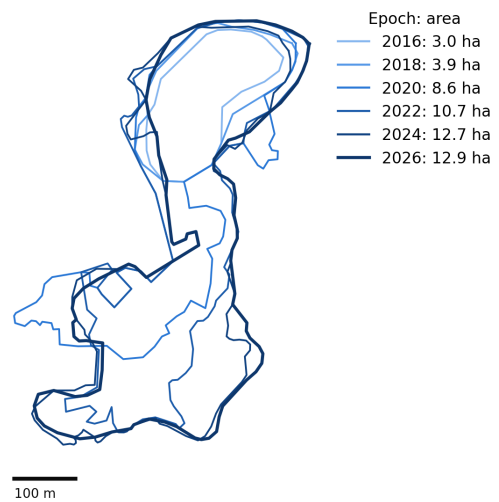
SUR-01 Surat LF-1 (Bhatar): 2.0 → 0.4 ha. Contracting. 2.0 → 0.4 ha.

Surat LF-2 (Khajod): traced outlines



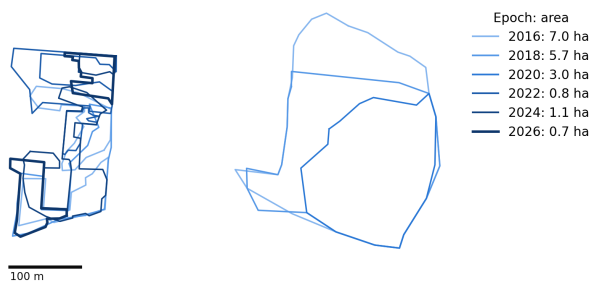
SUR-02 Surat LF-2 (Khajod): 68.7 → 31.2 ha. Cleared then regrowing. Part (a) cleared by 2020.

Vadodara LF-1: traced outlines



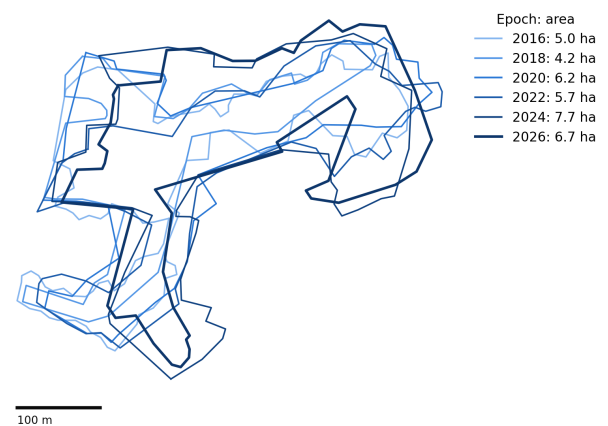
VAD-01 Vadodara LF-1: 3.0 → 12.9 ha. Expanding. 3.0 → 12.9 ha; strongest monotonic increase ($p = 0.003$).

Vadodara LF-2: traced outlines



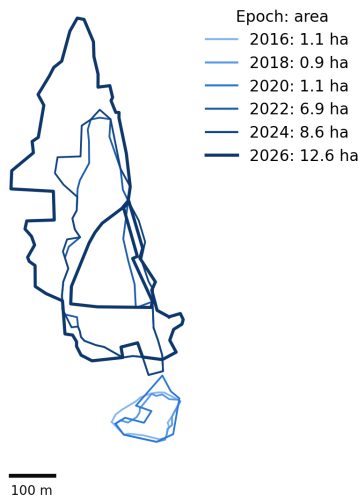
VAD-02 Vadodara LF-2: 7.0 → 0.7 ha. Contracting. Old dump gone by 2022; small patches since.

Bhuj LF-1: traced outlines

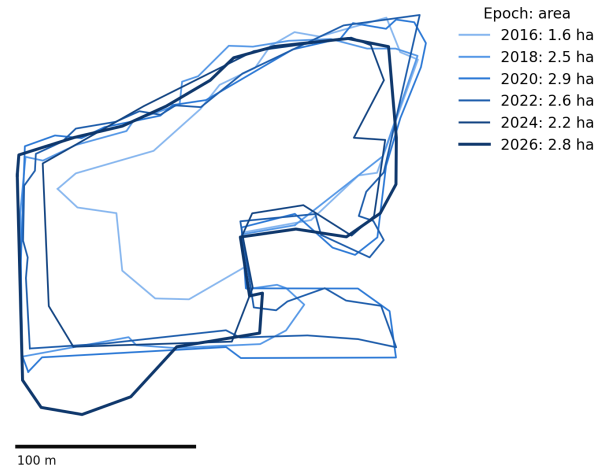


BHJ-01 Bhuj LF-1: 5.0 → 6.7 ha. Expanding (+34%). Irregular, spread footprint.

Gandhinagar LF-1: traced outlines



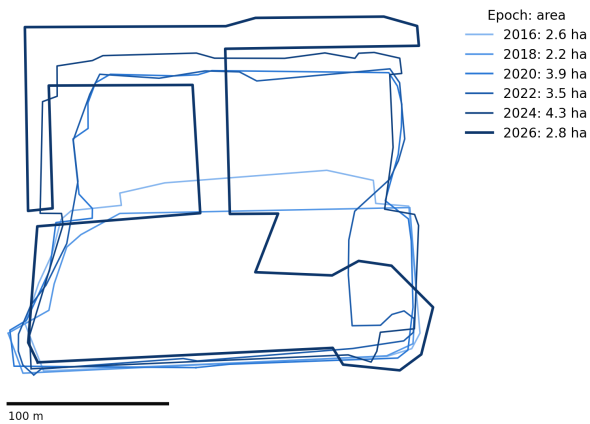
Jamnagar LF-1: traced outlines



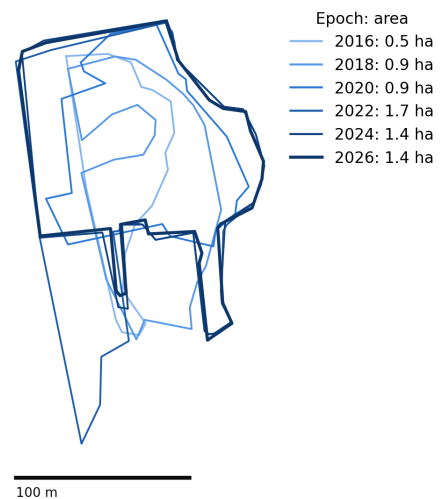
GNR-01 Gandhinagar LF-1: 1.1 → 12.6 ha. Expanding. 1.1 → 12.6 ha, most growth after 2020.

JAM-01 Jamnagar LF-1: 1.6 → 2.8 ha. Expanding (+80%), small site.

Mehsana LF-1: traced outlines



Vapi LF-1: traced outlines



MEH-01 Mehsana LF-1: 2.6 → 2.8 ha. Rise and fall. Parts cleared and built on; waste stacked.

VAP-01 Vapi LF-1: 0.5 → 1.4 ha. Expanding. Smallest site (0.5 → 1.4 ha).

Appendix B. Columns of the output tables

B.1 site_summary (one row per site)

Column	Meaning
site_id, site, city, group	Identifiers; group = Major city (five most populous) or Other city
n_parts_max, epochs	Largest number of separate polygons in any epoch; number of traced epochs (6)
area_2016_ha, area_2026_ha, net_change_ha, net_change_pct	First and last area and the change between them
peak_ha, peak_year, peak_to_2026_pct	Largest extent, its epoch, change from it to 2026
trough_ha, trough_year, trough_to_2026_pct	Smallest extent, its epoch, change from it to 2026
theil_sen_ha_per_yr	Theil-Sen slope of area
mk_S, mk_tau, mk_p_exact	Mann-Kendall S, tau-a, exact two-sided p (n = 6)
perimeter_2016_m, perimeter_2026_m, perimeter_change_pct, perimeter_theil_sen_m_per_yr	Perimeter (5 m simplified, parts summed) and its trend
shape_index_*, compactness_*, fractal_dim_* (2016, 2026)	Shape metrics (Section 6.1)
ever_under_waste_ha, newly_covered_since_2016_ha, no_longer_waste_2026_ha	Land turnover (Section 6.4)
significant_reversals	Direction changes among epoch-to-epoch changes larger than the threshold
mean_edge_uncertainty_pct, change_threshold_pct	Mean ± 2 m uncertainty (% of area) and the class threshold Th
trajectory	Class (Section 6.5)

B.2 Other tables

Table	Columns
site_by_epoch (84 rows)	site, site_id, epoch, status (traced / missing), n_parts, parts_traced (e.g. b+c), area_ha, area_unc_ha, perimeter_m, shape_index, compactness, fractal_dim
parts_by_epoch (103 rows)	site_id, site, epoch, part (main, a, b, c, a+b), area_ha, perimeter_m
transitions (70 rows)	site_id, from_epoch, to_epoch, gained_ha, lost_ha, persistent_ha, iou, d_area_ha, d_area_pct
city_totals (10 rows)	city, group, n_sites, 2016 ... 2026 (ha), net_change_ha, net_change_pct
group_totals (3 rows)	group (Major city, Other city, All sites), 2016 ... 2026 (ha), net_change_ha, net_change_pct
notes (Excel only)	Method notes and the full processing log (every placemark decision)

Appendix C. Parameters and fixed lists in the scripts

Name (script)	Value	Why
EPOCHS (analysis)	2016, 2018, 2020, 2022, 2024, 2026	Six two-yearly epochs
EPOCH_MAP (analysis)	2025 → 2024	A 2025 image stands in for 2024 where no usable 2024 image exists
CRS (analysis)	EPSG:32643, WGS 84 / UTM zone 43N	Metric CRS covering Gujarat (68-75° E)
SIMPLIFY (analysis)	5 m	Douglas-Peucker tolerance for perimeter only
BUF (analysis)	2 m	$\pm$ buffer for edge uncertainty
Threshold (analysis)	max(15%, 2 × mean edge uncertainty)	Class threshold
Overlap tolerance (analysis)	100 m ²	Parts overlapping more than this are logged
CITY, NAMES (analysis)	10 cities; 14 site names	Parsing and labels; extend for a new site
MAJOR (analysis)	AHM, SUR, VAD, RAJ, BHV	Five most populous cities
Ignored	Jamnagar LF-2	Remediated, not in the study
dpi (figures)	200	Slide-ready PNG

Name (script)	Value	Why
Colours (figures)	Major #2a78d6, other #eb6834; epoch ramp #86b6ef ... #0d366b	Colour-vision-safe pair; light = old, dark = new
Marker size, fig01 (figures)	40 + 18 × 2026 area (points ²)	Marker area grows linearly with waste area
Outline drawing (analysis → figures)	1 m simplification, coordinates relative to site centroid	Drawing only; areas use full outlines
SHORT, LOFF (figures)	Short site names; city label offsets on the map	Labels; extend for a new site